

Open Clinical Record

Software Requirements Specification

Version 2.0 — Week 8 Finalization Baseline

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Internship MVP — Patient Management, Patient Chart, Appointment Management

This document prevails over earlier drafts where they conflict.

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Open Clinical Record internship Minimum Viable Product (MVP). It is the authoritative requirements baseline for design validation, implementation, testing, demonstration, and formal evaluation.

Where earlier requirement drafts, research notes, or UI mocks conflict with this document, **this version prevails**.

1.2 Scope

Open Clinical Record is a focused **outpatient** electronic medical record supporting three business modules only:

1. **Patient Management** — registration, unique identification, search, demographic updates, and patient status (including deceased policy).
2. **Patient Chart** — longitudinal clinical context: allergies, medical history items, and **per-visit** documentation (vital signs, diagnoses, clinical notes), without overwriting prior attendances.
3. **Appointment Management** — scheduling, rescheduling, cancellation with reason, status transitions, check-in/queue, and appointment event history.

The product is **not** a full hospital EMR. Explicitly excluded: pharmacy dispensing, laboratory and imaging order management, billing and insurance, patient portal and native mobile apps, FHIR/HL7 external exchange, multi-facility enterprise scheduling, and AI clinical decision support.

1.3 Definitions

Term	Definition
MVP	Minimum Viable Product for the internship deliverable
Patient	Registered person with a unique medical record number (MRN)
Appointment	Scheduled (or walk-in typed) attendance slot with status lifecycle
Visit / ClinicalVisit	One facility attendance record holding clinical content
Draft	Editable visit status
Final	Closed visit; clinical content is immutable
JWT	JSON Web Token used for API authentication
RBAC	Role-based access control enforced at the API

1.4 References

Project scope, clinical domain rules, non-functional requirements, traceability matrix, access-control report, clinical visit model, technical documentation, and user guide in the project repository under docs/.

1.5 Document overview

Section 2 describes the product and users. Section 3 states functional requirements. Section 4 summarises non-functional requirements. Section 5 covers interfaces and data. Section 6 defines verification and acceptance. Section 7 lists exclusions.

2 Overall Description

2.1 Product perspective

OCR is a standalone web application with a clear three-tier structure:

```

Browser (React SPA + TypeScript)
  | REST / JSON + JWT
ASP.NET Core 8 Web API
  | EF Core 8 + Npgsql
PostgreSQL

```

Controllers remain thin; application services encapsulate workflows (patients, appointments, clinical chart, dashboard, audit).

2.2 Product functions

- Authenticate users and authorize operations by role at the API boundary.
- Register and manage patients, including deceased mark/clear policy.
- Maintain a longitudinal chart and per-visit clinical documentation.
- Schedule and manage appointments; check in patients; preserve history.
- Provide role-aware dashboards and basic operational statistics.
- Audit important actions for administrative review.

2.3 User characteristics — application roles

The MVP defines **exactly four** application roles:

Role	Claim	Primary responsibilities
Receptionist / Front Desk	Receptionist	Registration, booking, check-in, cancel/reschedule, mark deceased
Nurse / Clinical Staff	Nurse	Chart support, vitals, visit documentation (write)
Clinician / Doctor	Doctor	Chart review, clinical documentation, mark/clear deceased
System Administrator	Admin	Users, audit, configuration; operational access without default clinical authorship

Security principle: UI visibility is not security. The API shall enforce authorization independently of the client.

2.4 Constraints

- Internship time-box limits scope to the three modules above.
- PostgreSQL is the system of record; EF Core migrations are schema authority.
- Development training credentials must never be used in production.
- International interoperability standards (e.g. FHIR) are not MVP acceptance criteria.

2.5 Assumptions and dependencies

Users have a modern browser and network access to the API. PostgreSQL is available for non-test deployments. Appointment dates are treated as clinic-local calendar dates.

3 Functional Requirements

Priority **M** means mandatory for MVP acceptance. Identifiers align with the requirements traceability matrix.

3.1 Patient management (FR-PM)

ID	Requirement	Pri
FR-PM-001	Authorized users shall register a new patient with required demographics.	M
FR-PM-002	The system shall assign a unique medical record number to each patient.	M
FR-PM-003	Authenticated staff shall search patients by name and/or MRN.	M
FR-PM-004	Authorized users shall view a patient profile.	M
FR-PM-005	Authorized users shall update permitted demographic and contact fields.	M
FR-PM-006	The system should support duplicate awareness (e.g. search-before-create).	M
FR-PM-007	The system shall maintain patient status: Active, Inactive, or Deceased.	M
FR-PM-008	Marking a patient deceased (with provenance) shall be allowed for Admin, Doctor, and Receptionist — not Nurse.	M
FR-PM-009	Clearing deceased status shall be allowed for Admin and Doctor only.	M
FR-PM-010	The system shall reject new appointments for patients with status Deceased.	M

3.2 Patient chart and visits (FR-PC)

ID	Requirement	Pri
FR-PC-001	Authorized users shall open a patient chart in single-patient context.	M
FR-PC-002	The system shall display a chart summary (identity, status, key alerts).	M
FR-PC-003	Doctor and Nurse shall record and view allergies.	M
FR-PC-004	Doctor and Nurse shall record and view medical history items.	M
FR-PC-005	Clinical attendance shall be represented as a ClinicalVisit linked to the patient.	M
FR-PC-006	Visit statuses shall include Draft, Final, and Cancelled.	M
FR-PC-007	New visits shall default to Draft and remain editable until Final.	M
FR-PC-008	Final visits shall be immutable (no further clinical documentation edits).	M
FR-PC-009	Clinical staff shall record vital signs on a visit.	M
FR-PC-010	Clinical staff shall record diagnoses (primary/secondary) on a visit.	M
FR-PC-011	Clinical staff shall record clinical notes and plan/instructions on a visit.	M
FR-PC-012	Prior visits shall never be overwritten; a return creates a new visit.	M
FR-PC-013	Visit history shall be displayed chronologically on the chart.	M
FR-PC-014	Receptionist shall not create or edit clinical visit content (API forbidden).	M
FR-PC-015	Finalizing a visit linked to an appointment shall set that appointment to Completed.	M

3.3 Appointments (FR-AP)

ID	Requirement	Pri
FR-AP-001	Authorized staff shall create an appointment for an existing non-deceased patient.	M
FR-AP-002	Appointments shall be viewable for a selected date (list).	M
FR-AP-003	A calendar view (day/week) of appointments shall be supported.	M
FR-AP-004	Appointment types shall be supported as labels (Consultation, Follow-up, Walk-in, etc.).	M
FR-AP-005	Reschedule shall be allowed for Scheduled, Waiting, Cancelled, NoShow; blocked for CheckedIn, InProgress, Completed.	M
FR-AP-006	Cancel shall require a non-empty reason.	M
FR-AP-007	Statuses Scheduled, Waiting, CheckedIn, InProgress, Completed, Cancelled, NoShow shall have enforced transitions.	M
FR-AP-008	Check-in shall transition to CheckedIn and create a Draft visit when applicable.	M
FR-AP-009	Meaningful changes shall record AppointmentEvent history.	M
FR-AP-010	The active check-in queue shall exclude Completed, Cancelled, and NoShow.	M

ID	Requirement	Pri
FR-AP-011	Booking shall be available to Admin, Doctor, Nurse, and Receptionist.	M

3.4 Security, validation, and audit

ID	Requirement	Pri
FR-SEC-001	Users shall authenticate before accessing protected resources.	M
FR-SEC-002	Exactly four application roles shall be enforced as defined above.	M
FR-SEC-003	Unauthorized operations shall be rejected at the API (401/403).	M
FR-SEC-004	Important actions shall be auditable with actor and timestamp.	M
FR-SEC-005	Admin shall be able to list audit events.	M
FR-VAL-001	Required fields and relationships shall be validated before persistence.	M
FR-VAL-002	Patient and appointment business status rules shall be enforced.	M
FR-VAL-003	Error messages shall be clear without leaking sensitive internals.	M
FR-RPT-001	Dashboard statistics shall be available (e.g. today, waiting, checked-in).	M
FR-RPT-002	A simple reports view based on live stats may be provided (not a BI suite).	M

4 Non-Functional Requirements (Summary)

Detailed NFRs are maintained in `docs/03-requirements/non-functional-requirements.md`. MVP commitments include:

- **Security:** backend authorization, hashed passwords, least privilege, no secrets in source control.
- **Privacy:** minimum necessary access; synthetic data in demonstrations.
- **Integrity:** unique MRN; consistent relationships; history not silently deleted.
- **Performance:** common reads typically within approximately 2s and writes within approximately 3s under normal laboratory load; target of at least ten concurrent authenticated users.
- **Reliability:** failed operations must not report success; confirmed data preserved.
- **Usability:** role-aware navigation; visible patient context; distinguishable appointment statuses.
- **Auditability:** actor and timestamp; audit stream not editable through ordinary workflows.
- **Maintainability:** service-layer separation; migrations as schema source of truth.

5 Interfaces and Data

5.1 User interfaces

Web SPA with role-aware shell. Core screens: Login, Dashboard, Patients, Register, Chart, Appointments (list and calendar), Check-in queue, Profile, Admin audit.

5.2 Software interfaces

REST JSON API under `/api/`; PostgreSQL via EF Core 8.

5.3 Logical data

Patient (status, unique MRN); allergies and history items; ClinicalVisit with vital signs, diagnoses, notes; Appointment and AppointmentEvent; PatientDeathRecord; AuditEvent; identity tables for users and roles. **EF Core migrations** are authoritative; reference SQL under `docs/05-data/` is secondary.

6 Verification and Acceptance

6.1 Verification methods

- Automated API tests: access matrix, appointment transitions, clinical documentation/finalize, longitudinal visits.
- Manual end-to-end checklist under `docs/09-testing/`.
- Continuous integration: backend tests and frontend typecheck/build on `main`.

6.2 MVP acceptance criteria

1. Four roles can sign in with role-appropriate navigation.
2. A patient can be registered, searched, and opened in the chart.
3. An appointment can be booked, checked in, cancelled with reason, and rescheduled under allowed statuses.
4. Clinical staff can document a Draft visit and finalize; history shows multiple visits without overwrite.
5. Finalizing a visit completes the linked appointment for queue purposes.
6. Unauthorized clinical writes by Receptionist are rejected.
7. Deceased booking is rejected; mark/clear deceased follows the role matrix.
8. Technical documentation and user guide are available in the repository.

7 Exclusions

Prescription and pharmacy, laboratory and imaging workflows, billing, patient portal, FHIR exchange, multi-facility enterprise scheduling, and AI decision support are **not** MVP requirements.

Document control

Version 2.0 (27 September 2026). Implementation evidence is the `main` branch of the project repository. Subsequent changes shall increment the version and date.